Polar coordinate system

F18CE

Definition of Polar Coordinates

Let P be a point with cartesian coordinates (x, y).

- The line segment connecting the origin to P

  measures the distance from the origin to P has length r.
- The angle between the positive x-axis and

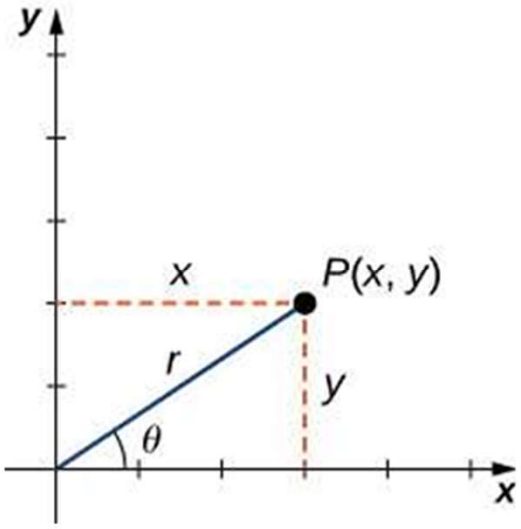
  the line segment has measure θ.

This correspondence is called the polar coordinate system.

It is clear that for all $(x,y) \in \mathbb{R}^2 \setminus \{(0,0)\}$,

$$\begin{cases} \cos\theta = \frac{x}{r} \\ \sin\theta = \frac{y}{r} \end{cases} \Rightarrow \begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases},$$

where r is called the radial coordinate and θ is the angular coordinate, and (r,θ) is the polar coordinate. Furthermore, $r^2 = x^2 + y^2$ and $\tan\theta = \frac{y}{x}$, where $0 < \theta < 2\pi$.



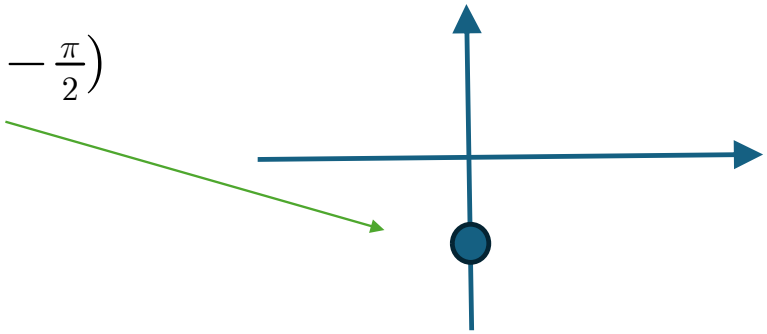
r, θ

- θ measures the angle anti-clockwise from the positive x-axis.
- θ is negative if it measures the angle clockwise from the positive x-axis.

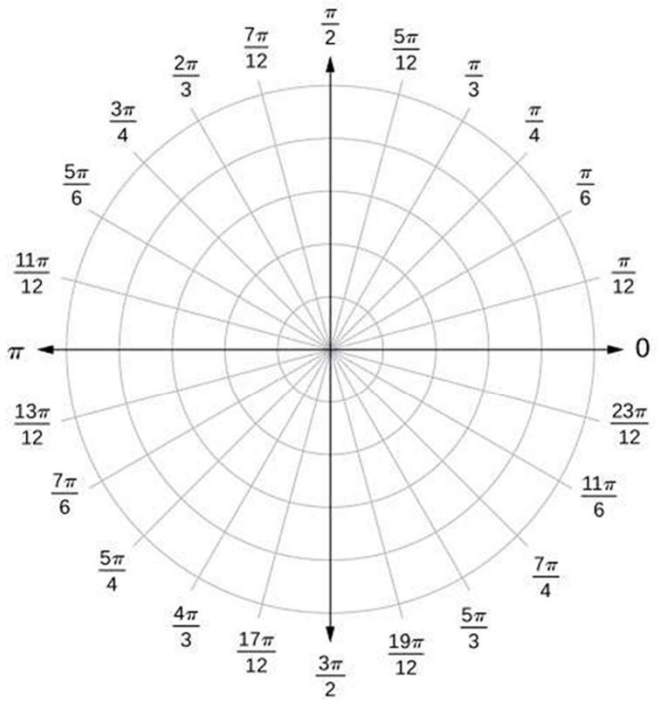
r, θ

- r is the directed distance measured from the origin based on the θ .
- The negative value of r is distance measured in the opposite direction of the distance from origin when θ is measured from the x-axis.

Example: $(1, \frac{3\pi}{2}) = (-1, \frac{\pi}{2}) = (1, -\frac{\pi}{2})$



The origin is called the pole



The origin is also called the pole,

and it is defined as $r = 0$

where θ is arbitrary.

If $r = 1$ and θ is arbitrary,

then we have a unit circle.

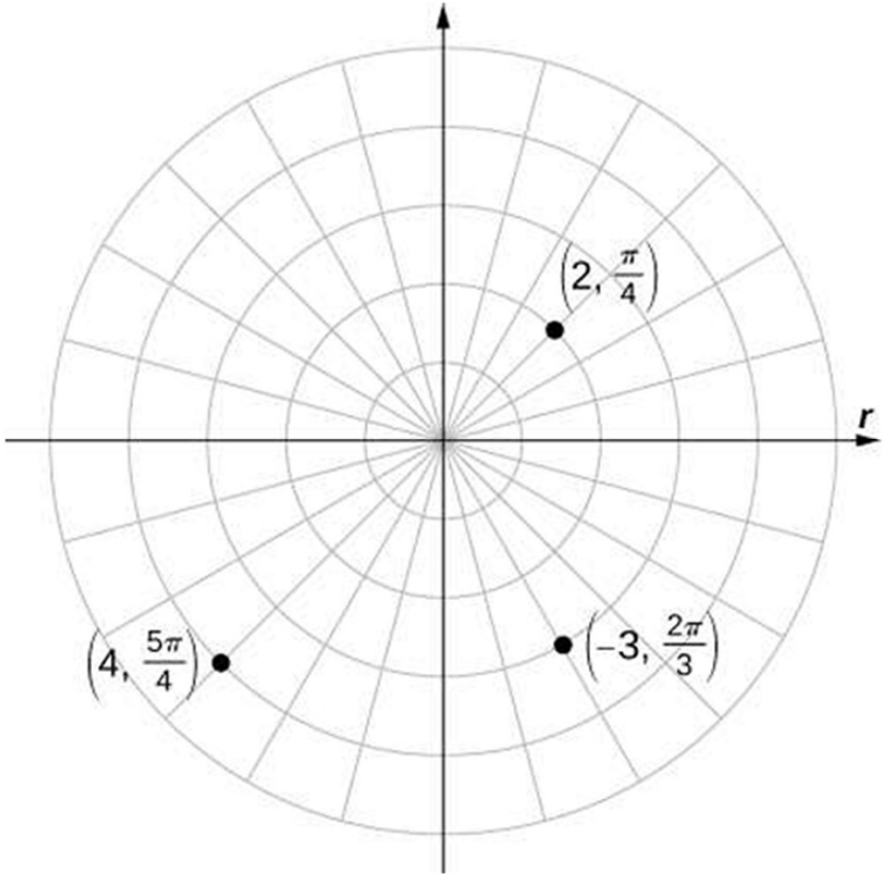
Example

Plot each of the following points on the polar plane.

(a) $(2, \frac{\pi}{4})$

(b) $(-3, \frac{2\pi}{3})$

(c) $(4, \frac{5\pi}{4})$



Polar curves

Now that we know how to plot points in the polar coordinate system, we can discuss how to plot curves. In the regular rectangular coordinate system (Cartesian system), we can graph a function $y = f(x)$ and create a curve in the Cartesian plane. In a similar fashion, we can graph a curve that is generated by function $r = f(\theta)$.

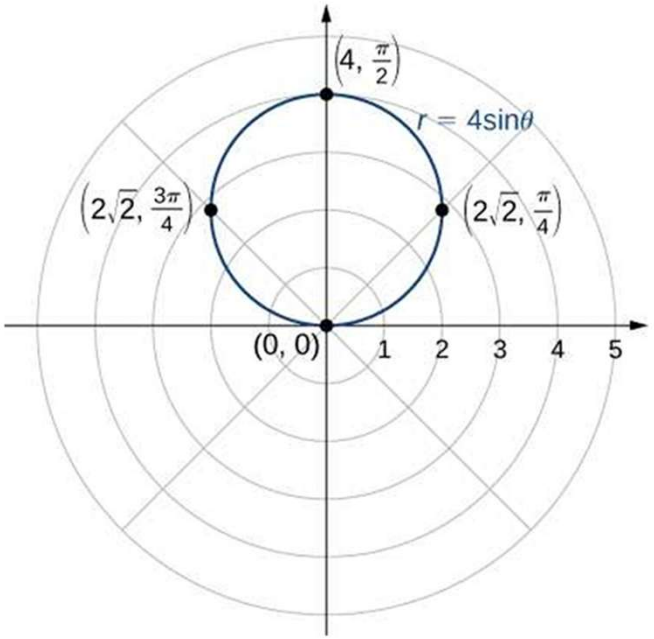
Plotting a curve in polar coordinates

1. Create a table, θ versus r.
2. For $\theta = \frac{n\pi}{12}$ or $\frac{n\pi}{6}$, $n \in \mathbb{N}$, find r.
3. Plot and connect each coordinate to look for a pattern.

Example

Example. Plot $r = 4\sin\theta$.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$
r	0	2	$2\sqrt{2}$	$2\sqrt{3}$	4	$2\sqrt{3}$	$2\sqrt{2}$	2
θ	π	$\pi + \frac{\pi}{6}$	$\pi + \frac{\pi}{3}$	$\pi + \frac{\pi}{4}$	$\pi + \frac{\pi}{2}$	$\pi + \frac{2\pi}{3}$	$\pi + \frac{3\pi}{4}$	$\pi + \frac{5\pi}{6}$
r	0	-2	$-2\sqrt{2}$	$-2\sqrt{3}$	-4	$-2\sqrt{3}$	$-2\sqrt{2}$	-2



We verify that

$$
\begin{aligned}
r = 4\sin\theta &\Rightarrow r^2 = 4r\sin\theta \\
&\Rightarrow x^2 + y^2 = 4y \\
&\Rightarrow x^2 + y^2 - 4y = 0 \\
&\Rightarrow x^2 + (y-2)^2 = 2^2
\end{aligned}
$$

which is a circle centred at $(0,2)$ with radius 2.

Transforming polar equation to rectangular coordinates

Identify the graph.

(a) $\theta = k$, k is fixed.

(b) $r = c$, c is fixed.

(c) $r = a\cos\theta + b\sin\theta$, a, b are fixed.

(a) $\tan^{-1}\frac{y}{x} = \theta = k \Rightarrow y = x\tan\theta = x\tan k\,.$

Since $\tan k$ is a constant, it follows that $\theta = k$

is a straight line passing the origin with slope $\tan k\,.$

(b) $r = c \Rightarrow r^2 = c^2 \Rightarrow x^2 + y^2 = c^2$.

This is a circle centred at origin with radius c.

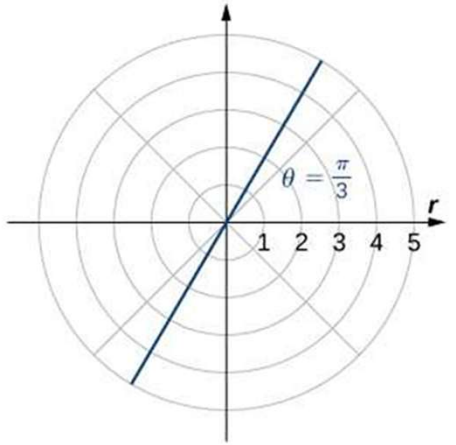
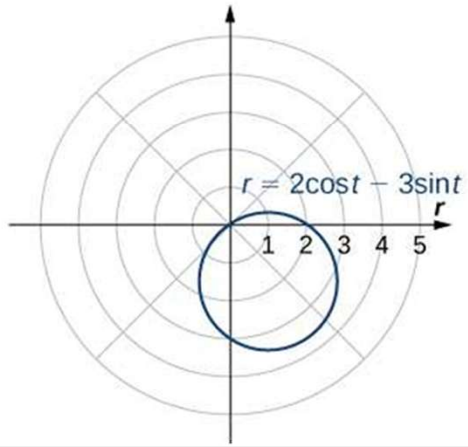
(c) $r = a\cos\theta + b\sin\theta$

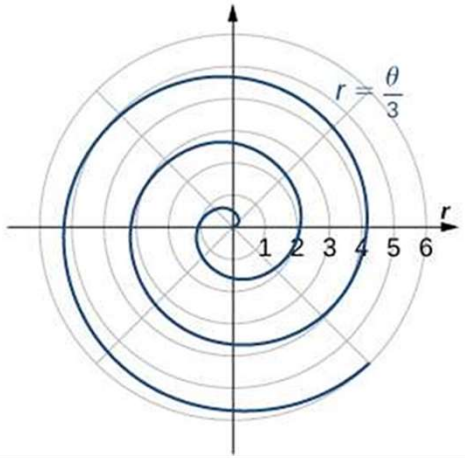
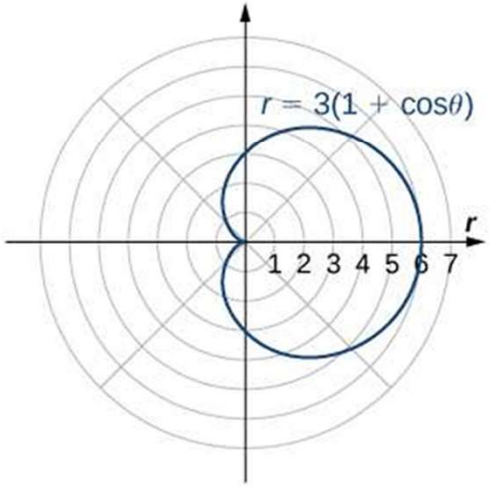
$\Rightarrow r^2 = r(a\cos\theta + b\sin\theta)$

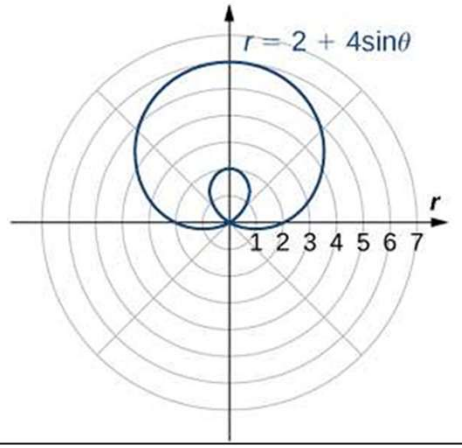
$\Rightarrow x^2 + y^2 = ax + by$

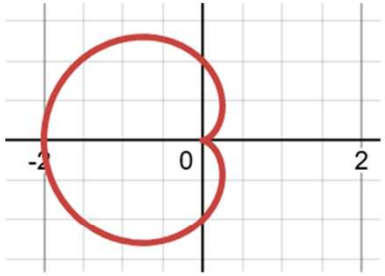
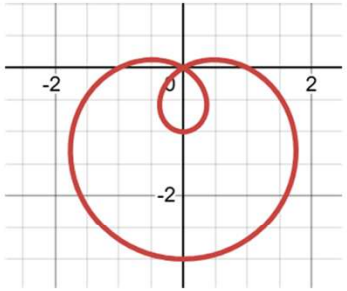
$\Rightarrow (x - \frac{a}{2})^2 + (y - \frac{b}{2})^2 = \frac{a^2+b^2}{4}$

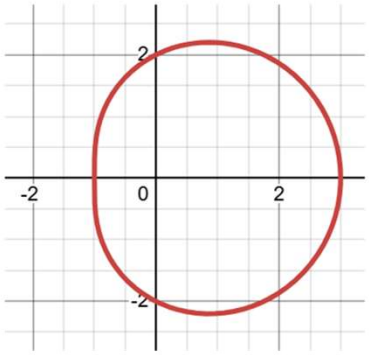
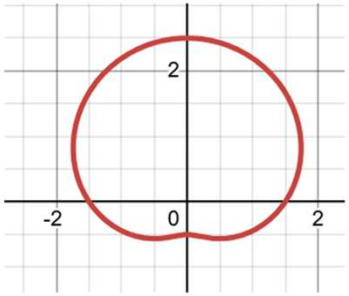
This is a circle centred at $(\frac{a}{2}, \frac{b}{2})$ with radius $\frac{\sqrt{a^2+b^2}}{2}$.

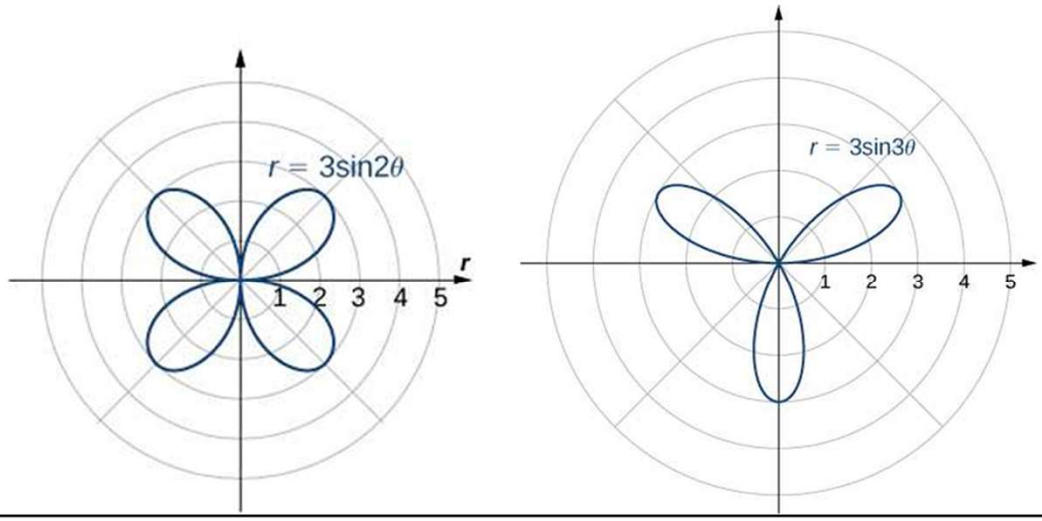
Name	Equation	Example
Line passing (0,0) with slope $\tan k$.	$\theta = k$, k is fixed.	
Circle	$r = c$, c is fixed, or $r = a\cos\theta + b\sin\theta$, a, b are fixed.	

Name	Equation	Example
Spiral	$r = a + b\theta$, a, b are fixed.	
Cardioid	$r = a(1 \pm \cos\theta)$ or $r = a(1 \pm \sin\theta)$, a is fixed	

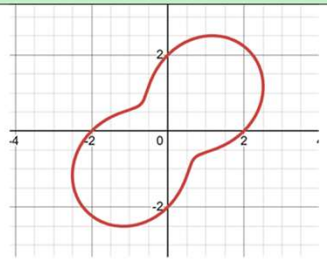
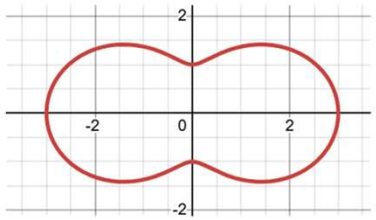
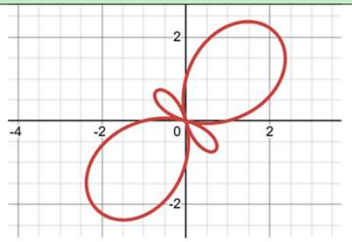
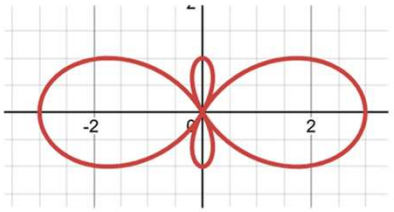
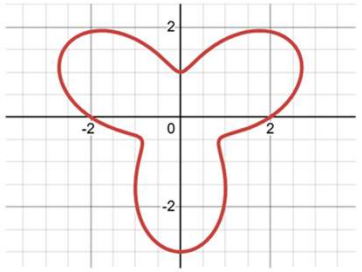
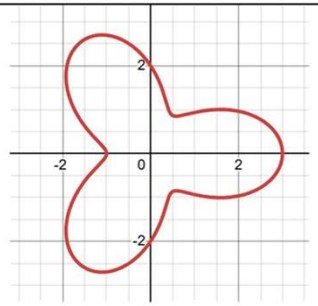
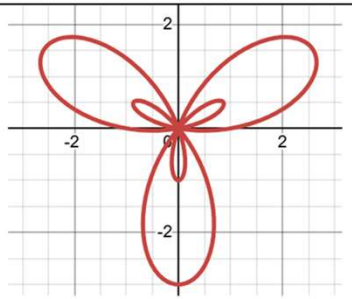
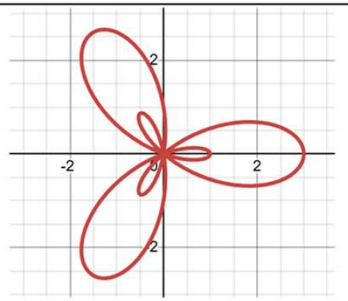
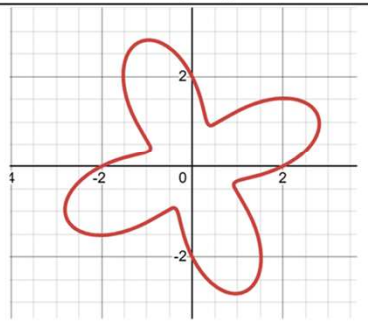
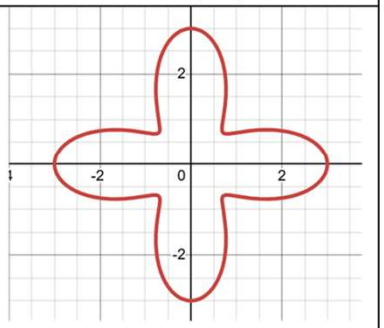
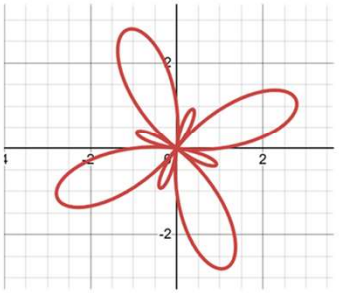
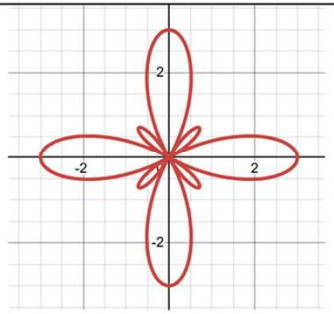
Name	Equation	Example
Limaçon (li-ma-song) *Ç or ç (C with cedilla, broken C) is a Latin script letter*	$r = a\cos\theta + b$, or $r = a\sin\theta + b$, a, b are fixed.	

Examples of limaçon (The $\pm$ sign of a decides the position of the shape)	
When $a = b$, $r = 1 - \cos\theta$.  Note that the function with cosine it has symmetry on the x-axis. This is called a cardioid (heart shape).	When $a < b$, $r = 1 - 2\sin\theta$.  Note that sine symmetric on the y-axis. This is called a limaçon with inner loop.

Examples of limaçon (The $\pm$ sign of a decides the position of the shape)	
When $a/b \geq 2$, $r = 2 + \cos\theta$.  Note that the function with cosine it has symmetry on the x-axis. This is called a convex limaçon.	When $2 > a/b > 1$, $r = \frac{3}{2} + \sin\theta$.  Note that the function with sine it has symmetry on the y-axis. This is called a dimpled limaçon.

Name	Equation	Example
Rose	$r = a\cos(b\theta)$, or $r = a\sin(b\theta)$	

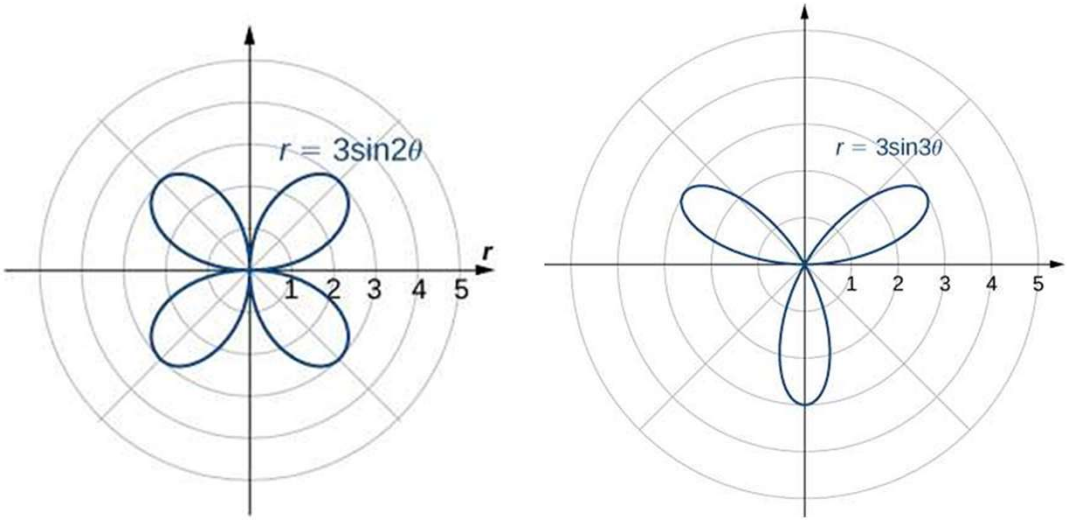
Lemniscate: $r = a\cos(b\theta) + c$ or $r = a\sin(b\theta) + c$, when $b \in \mathbb{N}$.

$a < c$	$b = 2$	 $r = \sin(2\theta) + 2$	 $r = \cos(2\theta) + 2$	$a > c$	$b = 2$	 $r = 2\sin(2\theta) + 1$	 $r = 2\cos(2\theta) + 1$
	$b = 3$	 $r = \sin(3\theta) + 2$	 $r = \cos(3\theta) + 2$		$b = 3$	 $r = 2\sin(3\theta) + 1$	 $r = 2\cos(3\theta) + 1$
	$b = 4$	 $r = \sin(4\theta) + 2$	 $r = \cos(4\theta) + 2$		$b = 4$	 $r = 2\sin(4\theta) + 1$	 $r = 2\cos(4\theta) + 1$

Remarks

1. A cardioid is a special case of a limaçon, in which $a = \pm b$ ($a = b$ or $a = -b$).

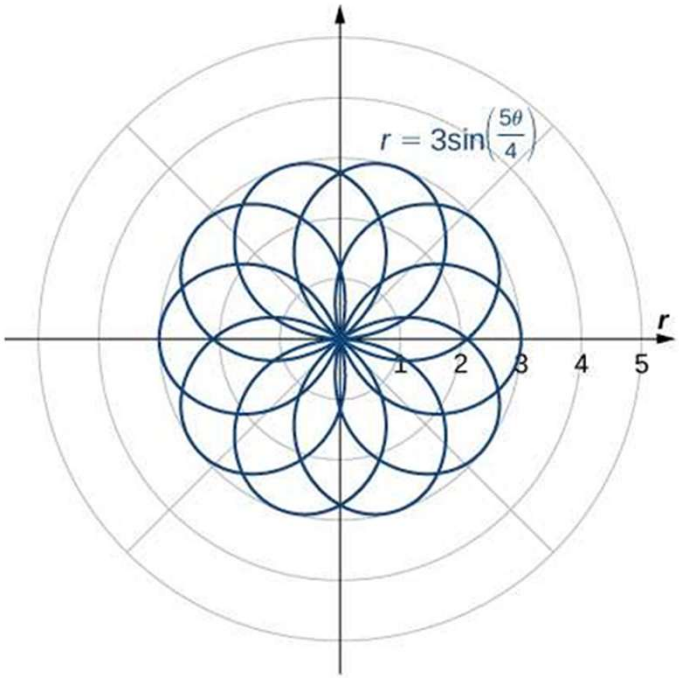
2. The rose is a very interesting curve. Note that $r = 3\sin 2\theta$ has four petals, but $r = \sin 3\theta$ has three petals. In general, if b is even, then there are $2b$ petals, while if b is odd, then there are b petals.

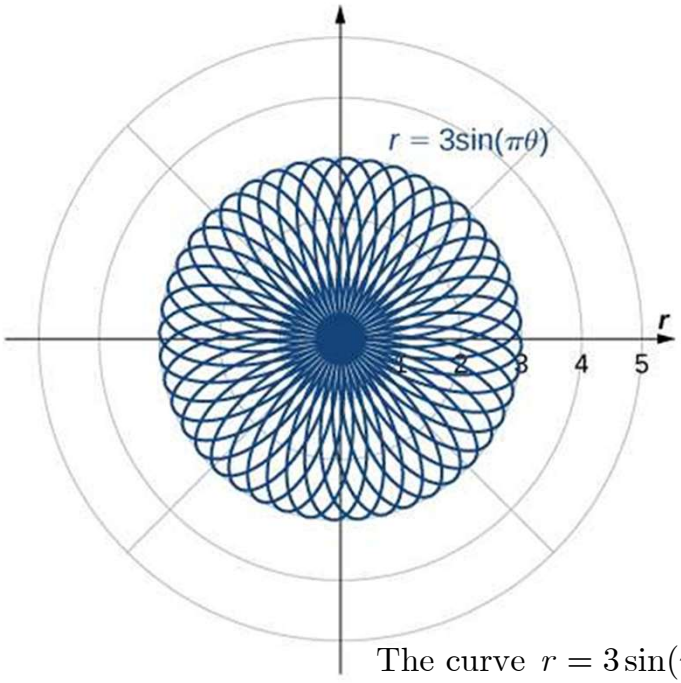


3. Consider the rose again, for $r = a\cos(b\theta)$, or $r = a\sin(b\theta)$,

(a) if $r \in \mathbb{Q}$, the curve is closed (i.e., it eventually ends where it started).

(b) if $r \in \mathbb{R} \setminus \mathbb{Q}$ (irrational), the curve never closed.





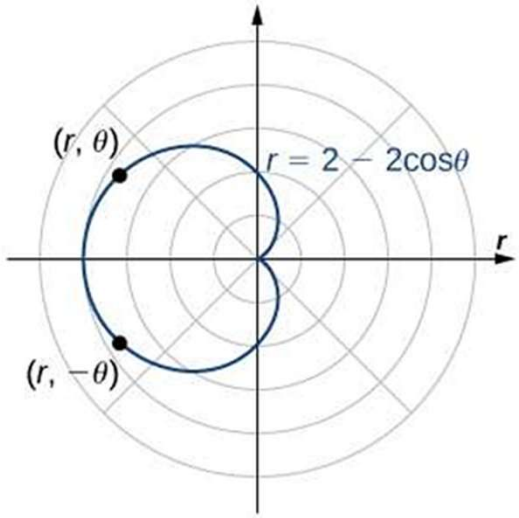
The curve $r = 3\sin(\pi\theta)$ never closes although it may appear it did, a close examination reveals that the petals just above the positive x-axis are slightly thicker. This is because the petal does not quite match up with the starting point!

Symmetry in Polar coordinates

Recall a function $f : D \to \mathbb{R}, D \subset \mathbb{R}$ is said to be an even function if $f(-x) = f(x)\ \forall x \in D$, and an odd function if $f(-x) = -f(x)\ \forall x \in D$. Geometrically, an even function is symmetric with respect to the y-axis (reflection on the y-axis), and an odd function is symmetric with respect to the origin (twice reflections on the y-axis and followed by the x-axis). Symmetry in polar curves works in a similar fashion.

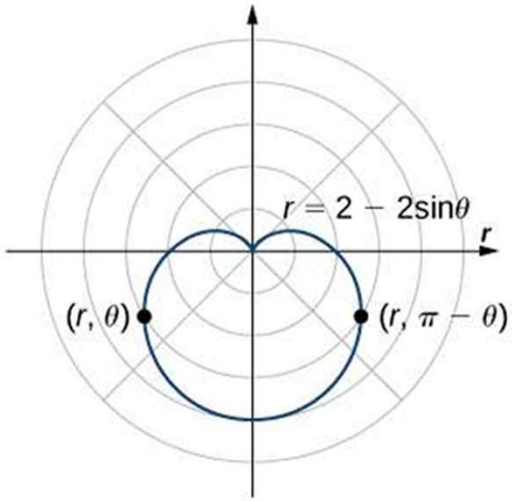
Symmetry with respect to the horizontal axis if

$r = f(\theta) = f(-\theta)$ for all (r, θ) on the graph.



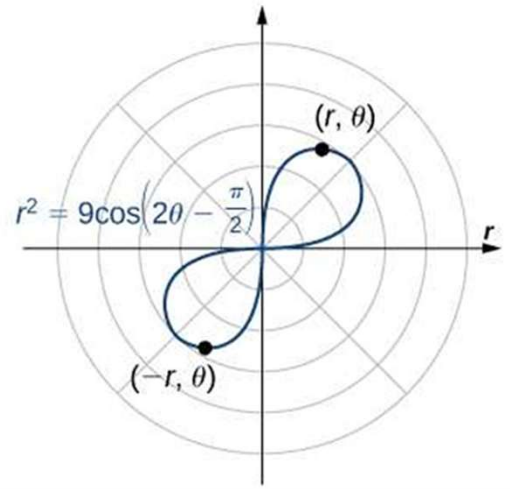
Symmetry with respect to the vertical axis if

$r = f(\theta) = f(\pi - \theta)$ for all (r, θ) on the graph.



Symmetry with respect to the pole if

$r = f(\theta) = f(\pi + \theta)$ for all (r, θ) on the graph.



Example

Example. Sketch the rose $r = 3\sin(2\theta)$.

Solution. Let $r = f(\theta) = 3\sin(2\theta)$.

First note that $f(-\theta) = -3\sin(2\theta) = -r \neq r$ and

$$\begin{aligned} f(\pi - \theta) &= 3\sin(2(\pi - \theta)) = 3\sin(2\pi - 2\theta) \\ &= 3\sin(-2\theta) = -3\sin(2\theta) \\ &= -r \neq r. \end{aligned}$$

So, the graph is not symmetry with respect to the vertical axis nor the horizontal axis.

Next, $f(\theta+\pi) = 3\sin(2(\theta+\pi))$.

$$\begin{aligned} &= 3\sin(2\theta+2\pi) \\ &= 3\sin 2\theta \\ &= r. \end{aligned}$$

So, the graph is symmetry with respect to the pole.

Note that $r = 3\sin 2\theta$ has four petals because 2 is even

and there are four petals, i.e., when $r = 0 \Rightarrow \theta = \frac{n\pi}{2}, n \in \mathbb{Z}$.

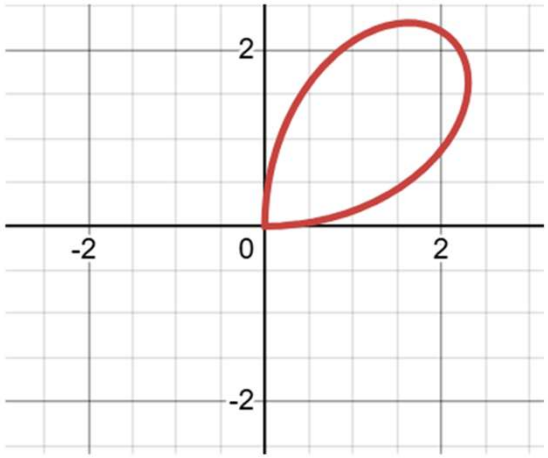
Also, since 2 is rational, the graph is closed (ends at where it starts).

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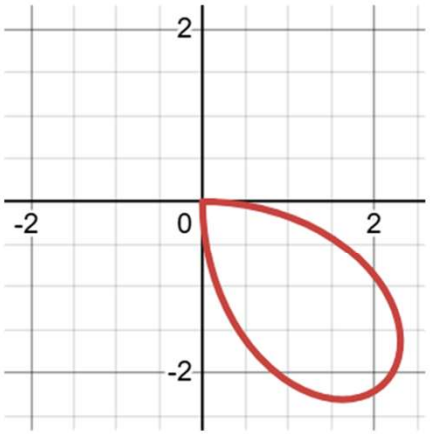
and there are four petals, i.e., when $r = 0 \Rightarrow \theta = \frac{n\pi}{2}, n \in \mathbb{Z}$.

Also, since 2 is rational, the graph is closed (ends at where it starts).

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$
r	0	$\frac{3\sqrt{3}}{2}$	3	$\frac{3\sqrt{3}}{2}$	0	$-\frac{3\sqrt{3}}{2}$	-3	$-\frac{3\sqrt{3}}{2}$

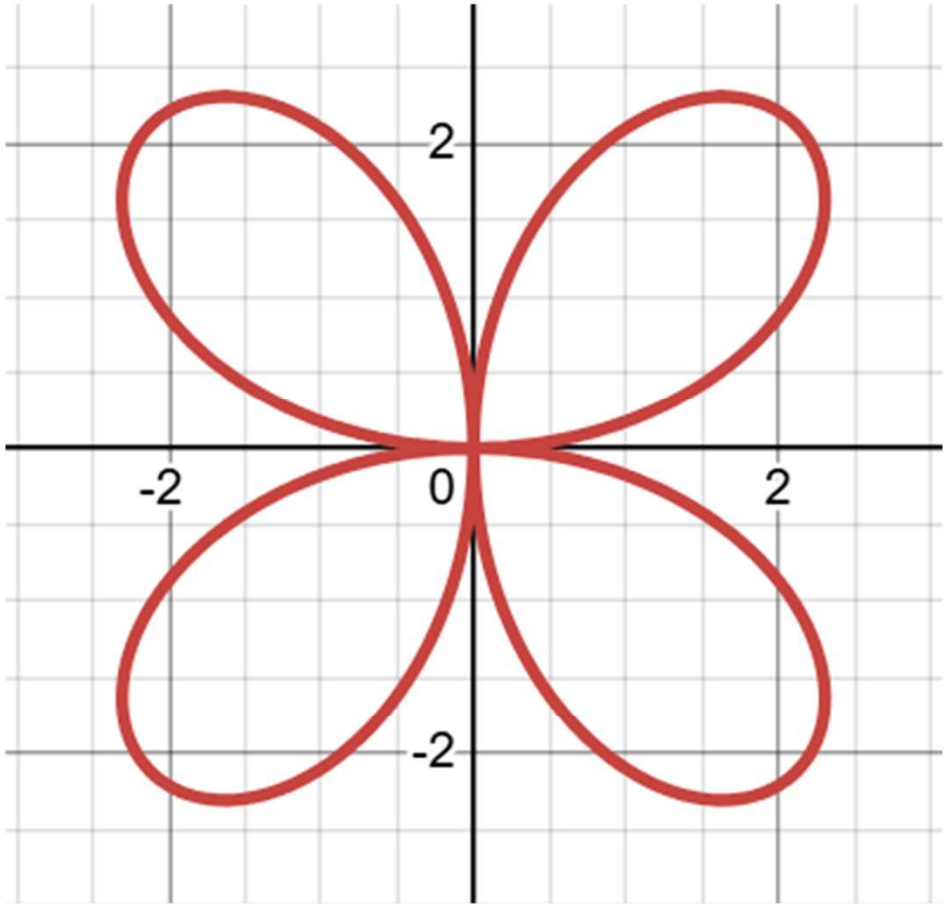


$0 \leq \theta \leq \frac{\pi}{2}$



$\frac{\pi}{2} \leq \theta \leq \pi$

Use the fact that the is symmetry with respect to the pole:



Exercise

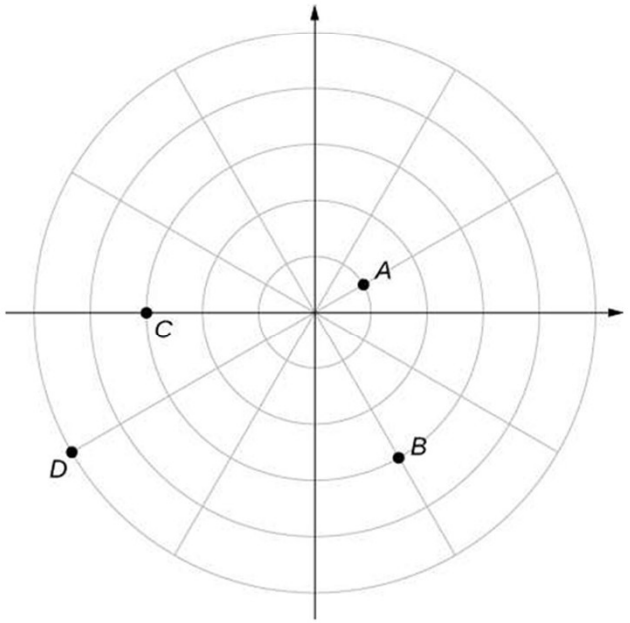
Show that the cardioid $r = a(1 \pm \cos\theta)$ is symmetric on the horizontal axis and $r = a(1 \pm \sin\theta)$ is symmetric on the vertical axis.

Exercise

1. Plot the points on a polar plane:

   $(3, \frac{\pi}{6})$, $(-2, \frac{5\pi}{3})$, $(0, \frac{7\pi}{6})$, $(-4, \frac{3\pi}{4})$

2. State the points on the polar plane.



3. Convert the cartesian coordinates to polar. $(2,2), (-6,8), (3,-4), (3,-\sqrt{3})$

4. Convert the polar coordinates to cartesian. $(-2, \frac{\pi}{6}), (0, \frac{\pi}{2}), (-3, \frac{3\pi}{4}), (-4.5, 6.5)$

5. Determine whether the following symmetry with respect to the x-axis, y-axis, and/or the pole.

(a) $r = 3\sin(2\theta)$ (b) $r^2 = 9\cos\theta$

(c) $r = \cos\frac{\theta}{5}$ (d) $r = 2\sec\theta$

(e) $r = 1 + \cos\theta$

6. Convert the following into rectangular equation.

(a) $r = 3$ (b) $\theta = \frac{\pi}{4}$ (c) $r = \sec\theta$ (d) $r = \csc\theta$

7. Convert the following into its polar form.

(a) $x^2 + y^2 = 16$ (b) $x^2 - y^2 = 16$ (c) $x = 8$

8. Convert the following into its polar form and sketch its graph.

(a) $3x - y = 2$ (b) $y^2 = 4x$

9. Convert the following into its rectangular form and sketch its graph.

(a) $r = 4\sin\theta$ (b) $r = 6\cos\theta$ (c) $r = \theta$ (d) $r = \cot\theta\csc\theta$

10. Sketch the following:

(a) $r = 1 + \sin\theta$ (b) $r = 3 - 2\cos\theta$ (c) $r = 2 - 2\sin\theta$

(d) $r = 5 - 4\sin\theta$ (e) $r = 3\sin(2\theta)$ (f) $r = 2\cos(3\theta)$

(g) $r = 3\cos\frac{\theta}{2}$ (h) $r^2 = 4\cos(2\theta)$ (i) $r^2 = 4\sin\theta$

(j) $r = 2\theta$

Remark

- I did not prepare the solutions. But if you did, please send me.
  I will type it and show to the future batches of students.
- I will write your name as the contributor to the works.